

Intergenerational Housing Allocation and Fertility

Part I: Compact Analytical Model

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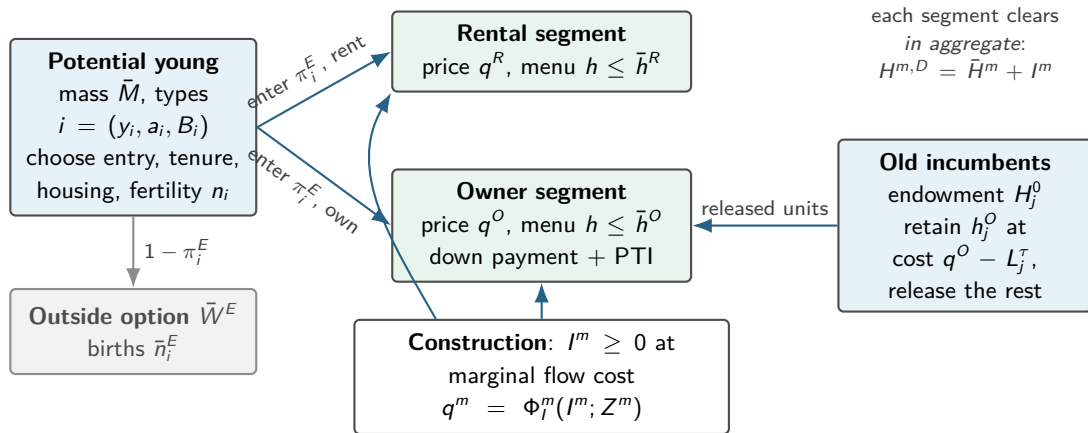
*When does limited access to family-capable housing reduce fertility,
and which policies actually relax that limit?*

The argument in one slide

1. **Children require space, not just goods.** Each child claims κ units of housing services and χ_i goods; parents consume what remains of the house and the budget.
2. **Family-sized space is rationed.** Rental menus stop at $\bar{h}^R$; buying requires a down payment and payment capacity. For each household, all of these frictions collapse into **one number**: the effective family-housing cap H_i^m .
3. **A binding cap makes children more expensive.** The shadow price of a child is $\chi_i + \kappa(q^m + \zeta_i^m)$: space for the child is priced at the household's *own* shadow value of space, which exceeds the market price by the access gap ζ_i^m .
4. **This yields a policy test.** Fertility responds only if the policy relaxes the *binding* component of H_i^m . The local effect is exposure $\times$ pass-through $\times$ fertility response, plus an entry margin.

One statistic does two jobs: under equal scaling $\Psi_i^m \propto \zeta_i^m$ — the gap that measures misallocation is the gap that predicts births.

The economy in one picture



No pairwise matching of old units to young buyers: (q^R, q^O) clear two aggregate markets. The asset price $P^O = q^O / \rho_p$ matters only for down payments.

Children require goods *and* space

Each child claims χ_i goods and κ units of housing services. Parents consume the residual:

$$c_i = C_i - \chi_i n_i, \quad s_i = h_i - \kappa n_i, \quad u_i^Y = \log c_i + \alpha \log s_i + \beta \log n_i.$$

- Same house, one more child $\Rightarrow$ less adult space s_i . Same budget, one more child $\Rightarrow$ less adult consumption c_i . **Children compete with their parents for space** — this is the engine of the model.
- Substituting $h = s + \kappa n$ into a branch budget $c + q^m h + \chi_i n = y_i$ gives

$$c_i + q^m s_i + \underbrace{(\chi_i + \kappa q^m)}_{\text{full price of a child}} n_i = y_i.$$

Unconstrained, a child costs its goods plus the space it occupies, at the *market* price of space.

- Because $\beta \log n_i$ rules out $n_i = 0$, every entrant has positive completed fertility: the model is about the **intensive margin**, not childlessness.

Family-sized space lives in the owner segment

Two segments, two menus. Renting and owning offer different size menus:

$$\mathcal{H}^R = \{h : 0 < h \leq \bar{h}^R\}, \quad \mathcal{H}^O = \{h : 0 < h \leq \bar{h}^O\}, \quad \bar{h}^R < \bar{h}^O.$$

Large family-sized units exist mainly in the owner segment. The caps are individual occupancy menus, not aggregate stocks.

Supply. Each segment has inherited stock plus net construction, supplied at increasing marginal cost:

$$H^m = \bar{H}^m + I^m, \quad q^m = \Phi_I^m(I^m; Z^m) \quad (\text{flow units}).$$

Buying needs finance. Owners face the asset price

$$P^O = \frac{q^O}{r + \delta^O + \tau^P},$$

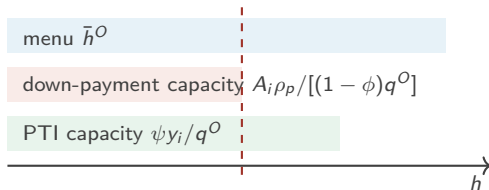
which enters *only* the down-payment constraint. Note for later: a higher property tax τ^P *lowers* P^O , hence lowers the required down payment.

Each household's frictions collapse into one cap

Conditional on entry, household i solves a renter branch and an owner branch and picks the better one, $W_i^Y = \max\{W_i^R, W_i^O\}$. In each branch, everything that limits family space is one number:

$$H_i^R = \bar{h}^R, \quad H_i^O = \min \left\{ \underbrace{\bar{h}^O}_{\text{menu}}, \underbrace{\frac{A_i(x) \rho_p}{(1-\phi) q^O}}_{\text{down payment}}, \underbrace{\frac{\psi y_i}{q^O}}_{\text{payment-to-income}} \right\}, \quad \rho_p = r + \delta^O + \tau^P.$$

H_i^O = shortest bar = **binding component**



Wealth is **pledgeable only**:

$$A_i(x) = a_i + xB_i + T_i^b(x), \quad x = 1 - \tau^b.$$

It backs the down payment but is not spendable in the flow budget. So inheritances and the estate tax work through *owner access*, not through consumption.

H_i^m is the only object a policy can usefully move in this model.

A binding cap shows up as an access gap ζ

When the cap binds, the household values space above its market price. The branch first-order conditions give

$$\frac{u_h}{u_c} = \frac{\alpha c_i^m}{s_i^m} = q^m + \zeta_i^m, \quad \zeta_i^m \geq 0,$$

where ζ_i^m is the **goods value of one more unit of space** — zero when unconstrained. For owners it decomposes as

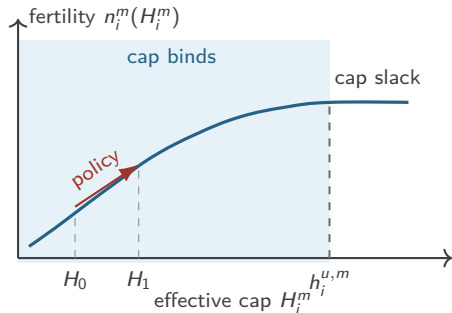
$$\zeta_i^O = \underbrace{\frac{\mu_i}{\lambda_i^O}(1 - \phi)P^O + \frac{\eta_i}{\lambda_i^O}q^O}_{\zeta_i^{O,F}: \text{ financial (DP, PTI)}} + \underbrace{\frac{\theta_i^O}{\lambda_i^O}}_{\text{menu}}.$$

Why this matters for fertility (Proposition 1)

$$\frac{\beta c_i^m}{n_i^m} = \chi_i + \kappa(q^m + \zeta_i^m)$$

A child's space requirement is priced at the household's *own* shadow value of space: children get more expensive even at unchanged market prices. ζ_i^m is the model's misallocation thermometer — it returns in the efficiency result *and* in the policy formula.

Space caps are fertility caps



With a binding cap, $h = H$ and fertility solves

$$\frac{\beta}{n} = \frac{\chi}{c(H)} + \frac{\alpha\kappa}{s(H)}.$$

Implicit differentiation:

$$\frac{dn}{dH} = \frac{\frac{\alpha\kappa}{s^2} - \frac{\chi q}{c^2}}{\frac{\beta}{n^2} + \frac{\chi^2}{c^2} + \frac{\alpha\kappa^2}{s^2}}.$$

Two forces: more space lowers the *crowding cost* of a child (numerator term $\alpha\kappa/s^2$), but the household also buys more housing at q , squeezing consumption ($\chi q/c^2$). The curve is within one tenure branch; each branch has its own curve.

When does more space mean more children?

Sign condition (exact, branch by branch)

Relaxing a strictly binding cap raises fertility at every binding cap level if and only if

$$\chi_i \leq \frac{\kappa q^m}{\alpha},$$

i.e. the child input bundle is **not more consumption-intensive** than the adult residual bundle.

- Read $\kappa q^m / \chi_i \geq \alpha$: the child's housing-to-goods expenditure ratio is at least the adult's. Children are, if anything, *space-intensive* — the empirically natural case.
- **Benchmark: equal scaling** $\chi_i = \kappa q^m / \alpha$. A child is then a scaled copy of the adult bundle, and any relaxation of a strictly binding cap raises fertility until the cap stops binding.
- If instead $\chi_i > \kappa q^m / \alpha$ (goods-hungry children), fertility can *fall* when the cap is relaxed near the unconstrained optimum: the extra housing purchase squeezes consumption, and consumption is what goods-hungry children need most.

One gap, two jobs: $\Psi_i^m \propto \zeta_i^m$

Define the within-branch fertility response $\Psi_i^m = \partial n_i^m / \partial H_i^m$ at a binding cap.

Corollary (equal scaling)

$$\Psi_i^m = \frac{\kappa}{\alpha} \frac{\zeta_i^m}{c_i^m} \Omega_i^m, \quad \Omega_i^m = \frac{\frac{\alpha}{s_i^m} + \frac{q^m}{c_i^m}}{\frac{\beta}{(n_i^m)^2} + \frac{\chi_i^2}{(c_i^m)^2} + \frac{\alpha \kappa^2}{(s_i^m)^2}} > 0.$$

- $\Psi_i^m > 0$ iff $\zeta_i^m > 0$: fertility responds exactly for the households whose access constraint bites, and more for those where it bites harder.
- The entry margin runs on the same object: relaxing the cap raises the inside value by $\Theta_i^m = \lambda_i^m \zeta_i^m = \zeta_i^m / c_i^m$ utils per unit.
- So **both** margins of the policy formula scale with the access gap per unit of adult consumption, ζ_i^m / c_i^m — the misallocation measure *is* the fertility lever.

The old side: retention can be tax-subsidized

Old incumbents enter with owner housing H_j^0 and choose how much to retain:

$$\max_{c,h,e} \log c + \gamma \log h + \mathcal{B}_j(e) \quad \text{s.t.} \quad c + e + (q^O - L_j^\tau)h = m_j + q^O H_j^0, \quad 0 \leq h \leq H_j^0.$$

- **Assessment lock-in:** $L_j^\tau = \tau^P P^O - \tilde{\tau}_j \tilde{P}_j \geq 0$ is the gap between the market property-tax cost and the protected assessed cost. Keeping the unit is artificially cheap; releasing it forfeits the protection.
- Interior retention satisfies $MV_j^O = q^O - L_j^\tau$: the old hold space past the point where their true valuation matches its social cost. L_j^τ is a **subsidy to sitting, not a preference**.
- Bequests are a true preference margin ($\mathcal{B}'_j = 1/c_j^O$), not a distortion. The planner counts old housing utility and warm glow — old occupancy is *not* inefficient by itself.

The young-access mechanism does not need this: it stands even at $L_j^\tau = 0$.
Lock-in is an optional *amplifier* of family-space scarcity.

Equilibrium: markets clear — clearing is not efficiency

A competitive equilibrium: prices (q^R, q^O) , asset price P^O , construction (I^R, I^O) such that

- young households solve both branches, pick tenure, enter with logit probability π_i^E ;
- old households solve the retention problem; builders set $q^m = \Phi_l^m$;
- both segments clear *in aggregate*:

$$\bar{M} \int \pi_i^E \mathbf{1}\{o_i=R\} h_i dG_Y = \bar{H}^R + I^R, \quad \bar{M} \int \pi_i^E \mathbf{1}\{o_i=O\} h_i dG_Y + \int h_j^O dF_O = \bar{H}^O + I^O;$$

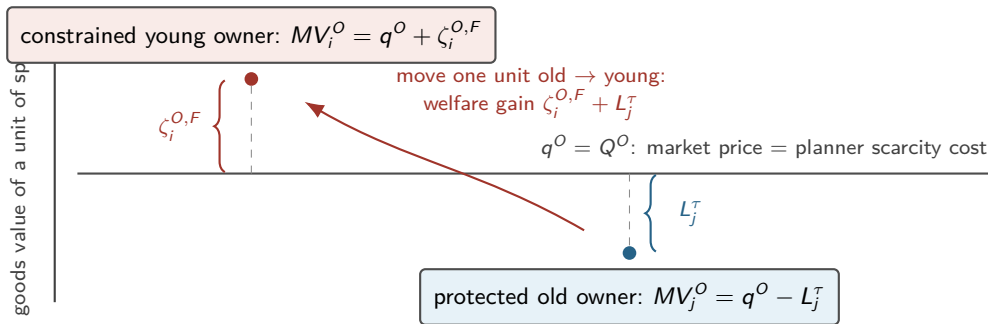
- housing payments are transfers, closed by lump-sum rebates; only construction uses real resources.

The punchline

Both segments can clear while the *marginal unit of owner space* is held by the wrong hands: financially constrained young value it at $q^O + \zeta_i^{O,F}$ and cannot buy more; tax-protected old hold it at $q^O - L_j^T$. Prices equate aggregates, not marginal values across people.

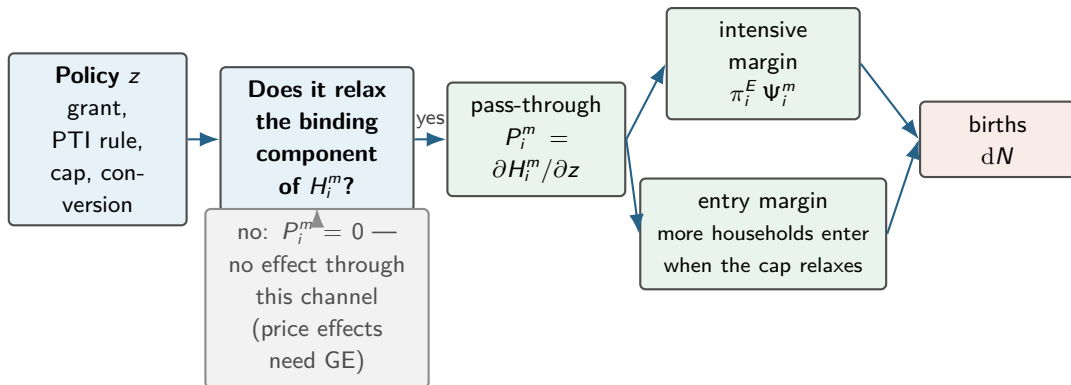
Two gaps separate the equilibrium from the planner

Planner: same preferences, same menus, same construction technology. Financial (DP/PTI) constraints are *implementation* constraints the planner can relax with financing instruments at zero marginal resource cost. Fertility-neutral ($\nu_i^g = 0$), conditional on entry measure and tenure assignments.



The equilibrium satisfies the planner's marginal conditions **only if** $\zeta_i^{O,F} = 0$ and $L_j^\tau = 0$. Binding *menus* are not a gap (the planner faces the same menus); entry and tenure-assignment margins are not characterized.

A policy test: does it relax the binding constraint?



The cap is a *minimum* of three components. A policy that improves a slack component does nothing here: a cheaper small apartment does not relax a binding down-payment constraint, and a down-payment grant does nothing for a renter capped by the rental menu.

The local policy decomposition

Holding prices, tenure choices, and active sets fixed ($\mathcal{B}_m$ = entrant types whose cap binds in branch m):

$$\left. \frac{dN}{dz} \right|_{q,o,\mathcal{A}} = \bar{M} \sum_{m \in \{R,O\}} \int_{\mathcal{B}_m} \left[\underbrace{\pi_i^E \Psi_i^m}_{\text{intensive: entered mass} \times \text{fertility response}} + \underbrace{(n_i^m - \bar{n}_i^E) \frac{\pi_i^E (1 - \pi_i^E)}{\kappa_E} \Theta_i^m}_{\text{entry: births per entrant} \times \text{sensitivity} \times \text{value of relaxation}} \right] \underbrace{P_i^m(z)}_{\text{pass-through to the cap}} dG_Y(i)$$

Read it as: exposure $\times$ pass-through $\times$ response, plus entry. Births count relative to outside-option fertility $\bar{n}_i^E$ (zero under the baseline normalization).

What this formula deliberately leaves out

Price and construction responses, fiscal costs Ξ^Π , active-set changes — and **tenure switching**, which is first order when types sit near the indifference boundary $W_i^R = W_i^O$ and fertility jumps across branches. The formula is the fixed-tenure component; the quantitative model evaluates the switching margin.

Which policies pass the test

Pass (when the named component binds)

- **Down-payment grants / LTV relief** ($A_i \uparrow$, $\phi \uparrow$): $P_i^O = \rho_p / [(1 - \phi)q^O] > 0$ for DP-constrained owners.
- **PTI relaxation** ($\psi \uparrow$): $P_i^O = y_i / q^O > 0$ for payment-constrained owners.
- **Family-sized rental supply / conversion** ($\bar{h}^R \uparrow$): $P_i^R = 1$ for menu-capped renters.
- **Assessment reform** (reassessment): shrinks L_j^τ , releases old-held space — works through the old margin.

No effect through this channel

- Subsidies to units *below* the binding size: the binding component does not move.
- Demand subsidies that change prices but not H_i^m : price effects exist but require the GE analysis, not this formula.
- Liquidity for households whose constraints are slack: pledgeable wealth has no income effect by construction.

Two-sided instrument. The property tax: a higher τ^p lowers P^O and relaxes young down payments, while assessment *protection* is what creates old lock-in. Level and protection are separate levers.

Takeaways

1. **Family-housing access is a fertility margin.** Children are priced at the household's own shadow value of space: $\beta c_i/n_i = \chi_i + \kappa(q^m + \zeta_i^m)$. If the child bundle is not more goods-intensive than the adult's ($\chi_i \leq \kappa q^m/\alpha$), relaxing a binding cap raises fertility.
2. **Two measurable gaps separate equilibrium from the planner:** the young financial access gap $\zeta_i^{O,F}$ and the old assessment advantage L_j^τ . Moving one marginal unit of owner space from a protected incumbent to a constrained young family is worth $\zeta_i^{O,F} + L_j^\tau$. The young mechanism stands without the old one.
3. **Policy works only through the binding component of H_i^m .** Local effect = exposure $\times$ pass-through $\times$ response + entry; under equal scaling $\Psi_i^m \propto \zeta_i^m$, so the policies that raise births most are the ones aimed at the largest access gaps.

Appendix: the branch derivative

If $0 < H < h^u$, the cap binds, $h(H) = H$, and

$$c(H) = y - qH - \chi n(H), \quad s(H) = H - \kappa n(H).$$

Fertility solves $F(n, H) = \frac{\beta}{n} - \frac{\chi}{c(H)} - \frac{\alpha\kappa}{s(H)} = 0$, and

$$F_n = -\left[\frac{\beta}{n^2} + \frac{\chi^2}{c^2} + \frac{\alpha\kappa^2}{s^2}\right] < 0, \quad F_H = \frac{\alpha\kappa}{s^2} - \frac{\chi q}{c^2},$$

$$\frac{dn}{dH} = -\frac{F_H}{F_n} = \frac{\frac{\alpha\kappa}{s^2} - \frac{\chi q}{c^2}}{\frac{\beta}{n^2} + \frac{\chi^2}{c^2} + \frac{\alpha\kappa^2}{s^2}}.$$

Unconstrained benchmark (Cobb–Douglas in (c, s, n) at prices $(1, q, \chi + \kappa q)$):

$$c^u = \frac{y}{1 + \alpha + \beta}, \quad n^u = \frac{\beta y}{(1 + \alpha + \beta)(\chi + \kappa q)}, \quad \frac{qs^u}{c^u} = \alpha.$$

Appendix: equal scaling and the access gap

At a binding cap, $\alpha c_i^m / s_i^m = q^m + \zeta_i^m$. Under equal scaling $\chi_i = \kappa q^m / \alpha$, the numerator of Ψ_i^m factors:

$$\frac{\alpha \kappa}{(s_i^m)^2} - \frac{\chi_i q^m}{(c_i^m)^2} = \frac{\kappa}{\alpha} \left(\frac{\alpha}{s_i^m} - \frac{q^m}{c_i^m} \right) \left(\frac{\alpha}{s_i^m} + \frac{q^m}{c_i^m} \right), \quad \frac{\alpha}{s_i^m} - \frac{q^m}{c_i^m} = \frac{\zeta_i^m}{c_i^m},$$

so

$$\Psi_i^m = \frac{\kappa}{\alpha} \frac{\zeta_i^m}{c_i^m} \Omega_i^m, \quad \Omega_i^m = \frac{\frac{\alpha}{s_i^m} + \frac{q^m}{c_i^m}}{\frac{\beta}{(n_i^m)^2} + \frac{\chi_i^2}{(c_i^m)^2} + \frac{\alpha \kappa^2}{(s_i^m)^2}} > 0.$$

The envelope value of relaxing the cap is $\Theta_i^m = \partial W_i^m / \partial H_i^m = \lambda_i^m \zeta_i^m$ with $\lambda_i^m = 1/c_i^m$, and the logit entry response is $\partial \pi_i^E / \partial W_i^Y = \pi_i^E (1 - \pi_i^E) / \kappa_E$.

Appendix: planner conditions and the comparison

With goods multiplier Λ and segment multipliers ΛQ^m , interior planner allocations satisfy, away from menu corners,

$$\frac{u_{h,i}^Y}{u_{c,i}^Y} = Q^m, \quad \frac{u_{h,j}^O}{u_{c,j}^O} = Q^O, \quad Q^m = \Phi_I^m(I^m; Z^m), \quad \frac{\beta c_i}{n_i} + \nu_i^g = \chi_i + \kappa Q_i^{o^P}.$$

Endogenous logit entry enters all allocation FOCs through a common factor that cancels from these ratios.

At an interior competitive allocation, clearing and competitive construction give $Q^R = q^R$ and $Q^O = q^O$. Competitive marginal values are

$$MV_i^R = q^R + \zeta_i^R, \quad MV_i^O = q^O + \zeta_i^O, \quad MV_j^O = q^O - L_j^\tau.$$

If DP/PTI are relaxable at zero marginal resource cost, the CE satisfies the planner's conditional FOCs only if $\zeta_i^{O,F} = 0$ and $L_j^\tau = 0$; menus common to both problems generate no gap. With positive instrument costs, compare the gap to the marginal cost. Entry and tenure-assignment efficiency are not characterized.

Appendix: accounting conventions

- **Flow normalization.** q^m is both the user cost and the marginal resource cost of segment- m services; Φ^m is in flow units, so $q^m = \Phi_I^m$. P^O only converts q^O into the purchase price for down payments. (Under asset-unit costs, $P^O = \Phi_I^O$ and the property tax would also distort construction.)
- **Stocks.** $\int H_j^0 dF_O + H_{\text{pass}}^O = \bar{H}^O$; rental stock held by passive landlords; all passive and fiscal income rebated lump-sum.
- **Bequest loop is open (reduced form).** Old bequest expenditure e_j is a current resource use; young exposure B_i is a primitive of G_Y . Hence the estate tax τ^b works only through pledgeable resources $A_i(x)$, with no old-side bequest distortion.
- **Outside fertility.** N counts births among entrants; with outside births $\bar{n}_i^E$, total fertility adds $(1 - \pi_i^E)\bar{n}_i^E$ and entry-margin terms use $n_i^Y - \bar{n}_i^E$.